Part II

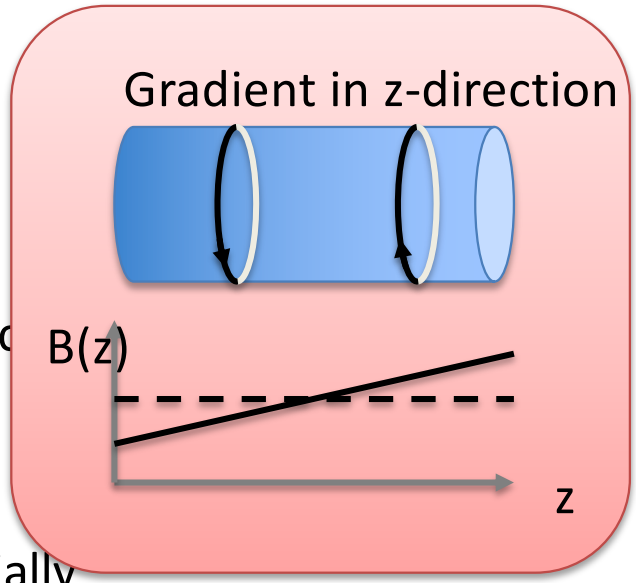
Recap

- Static magnetic field (B_0)
  - Net magnetization precesses with Larmor frequency
- Varying magnetic field with Larmor frequency (B_1, “RF pulse”)
  - Rotates the magnetization of all spins on resonance
  - Excites signal
  - Refocuses static magnetic field variations
- Sequence timing (TE and TR) controls image contrast depending on T_1 and T_2 relaxation

- Spatial localization??

Magnetic field gradients

- Using coils, an additional magnetic field can be superimposed over the B_0 field
  - Field strength varies
  - Field direction *always* in z-direction
- The current can be varied in the gradient coils
  - One coil set in each direction x, y and z
  - Combined will create a gradient in arbitrary direc
- Larmor frequency is proportional to magnetic field
- Using gradients makes Larmor frequency vary spatially



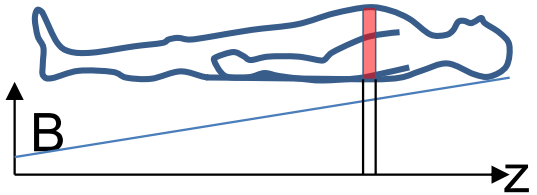
Spatial localization of signal

Use magnetic field gradients to modify the local field strength

- Larmor frequency changes linearly with space

Combine RF pulse with gradient on

- Excites one slice, leaving rest of body unaffected

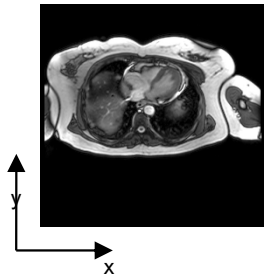


$$f = \gamma(B_0 + G_z z)$$

Read signal with in-plane encoding

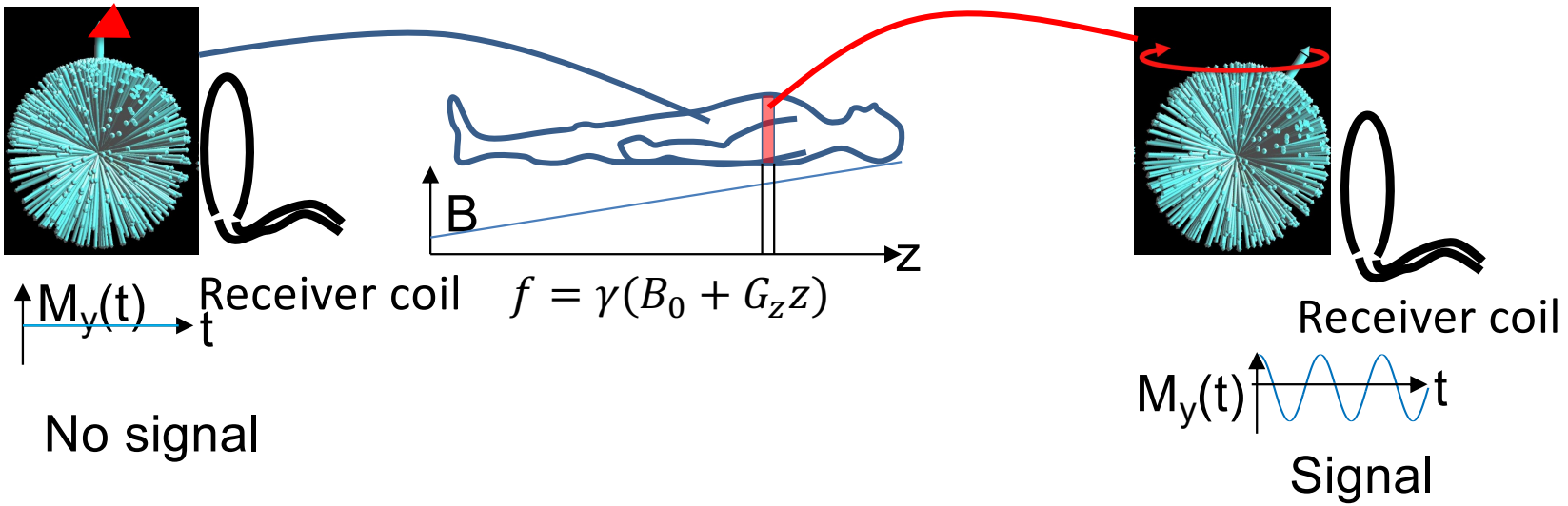
- Signals can be separated by frequency
- *k*-space encoding

$$f = \gamma(B_0 + G_x x)$$

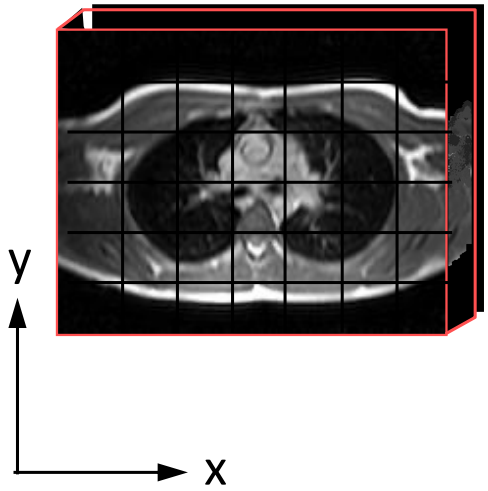


Slice selection

- With a gradient, the Larmor frequency varies linearly with space
- Only the spins in resonance will be excited and thus provide signal
- Use a non-zero bandwidth to excite a slice



2D in-plane encoding

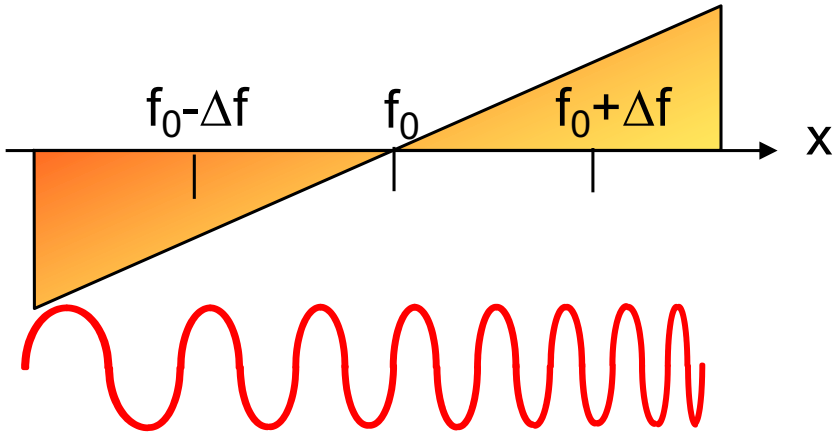


- The signal received comes from the whole excited volume
- We use gradients in two steps to encode the x and y directions
- Frequency encoding (convention: x)
- Phase encoding (convention: y)

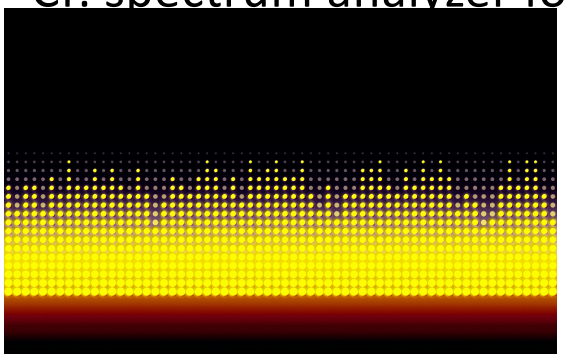
Frequency encoding direction (x)

- A gradient is turned on during signal reception
- Signals from different positions have different frequencies

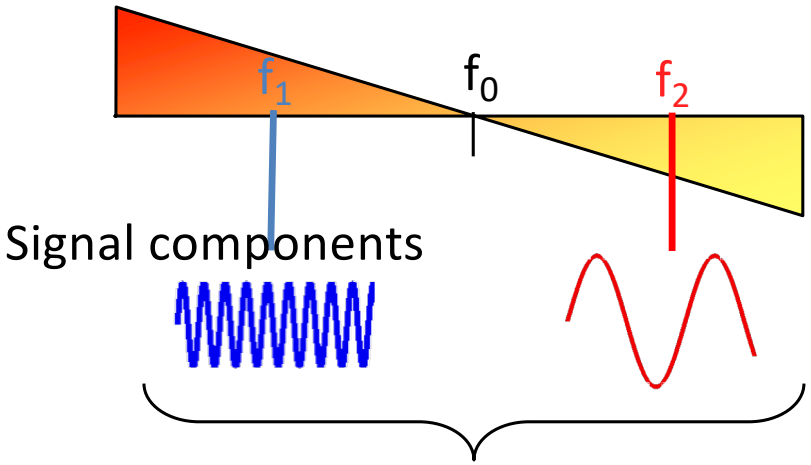
$$f = \gamma(B_0 + G_x x)$$



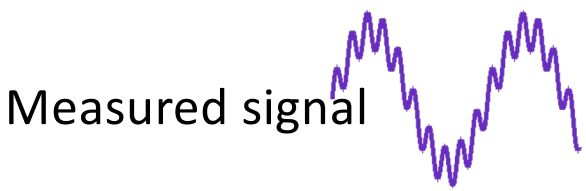
Cf: spectrum analyzer for music

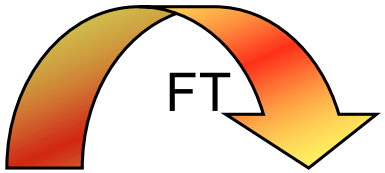


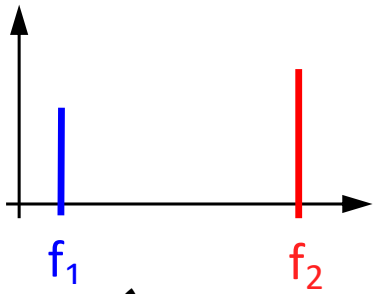
The Fourier transform



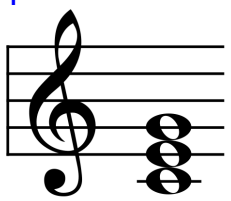
The Fourier transform recovers the frequencies and separates into f_1 and f_2



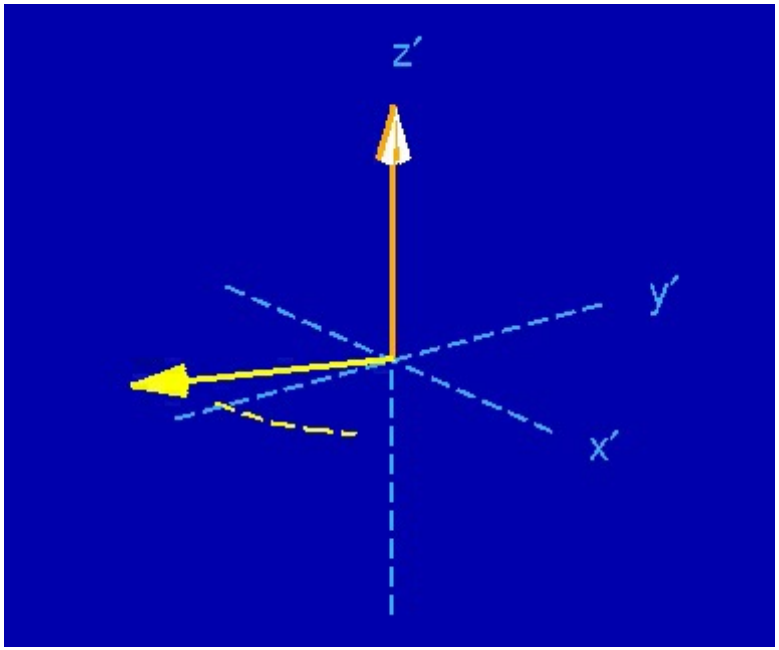




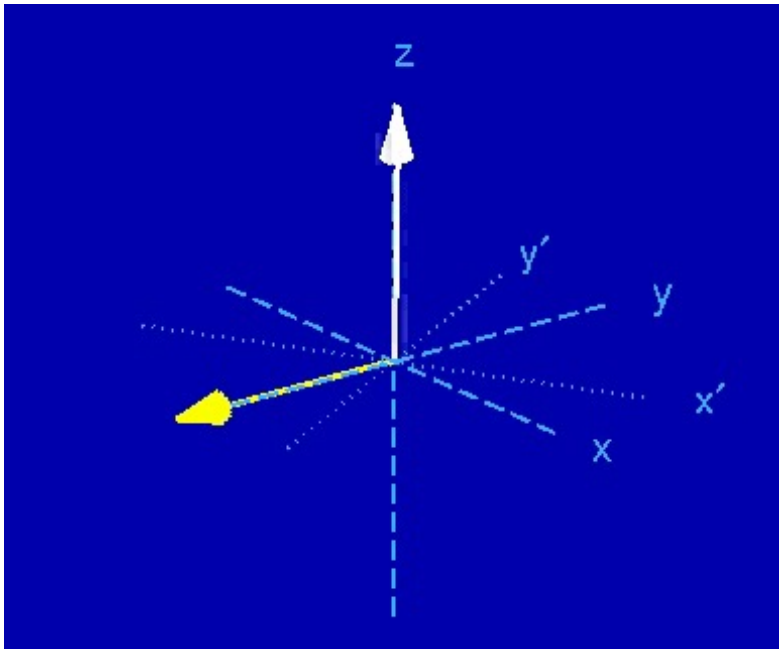
cf: Music chords



Rotating frame of reference



Static frame of reference



Rotating frame of reference

Movies from http://mrsrl.stanford.edu/~brian/intromr/ (Brian Hargreaves)

Frequency encoding

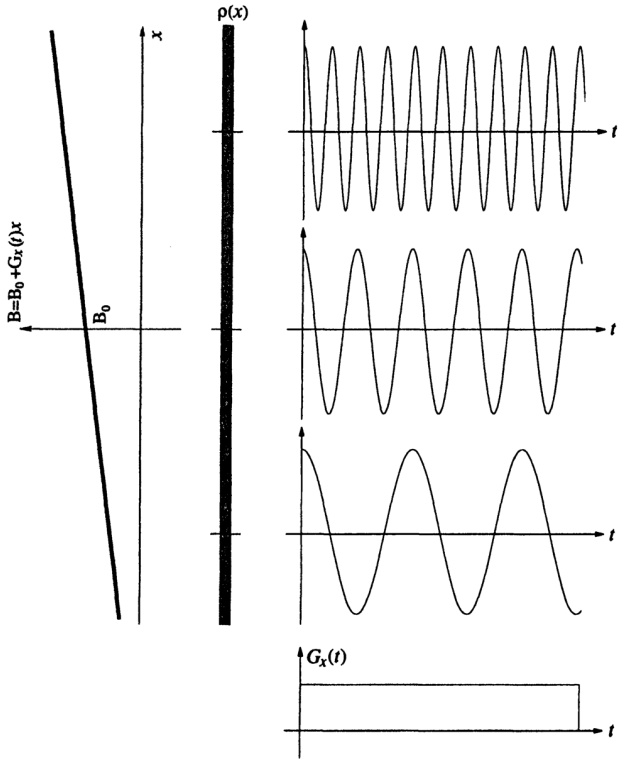


Figure 5.7 Localized signals from a hypothetical one-dimensional object in the presence of a frequency-encoding gradient.

Phase encoding

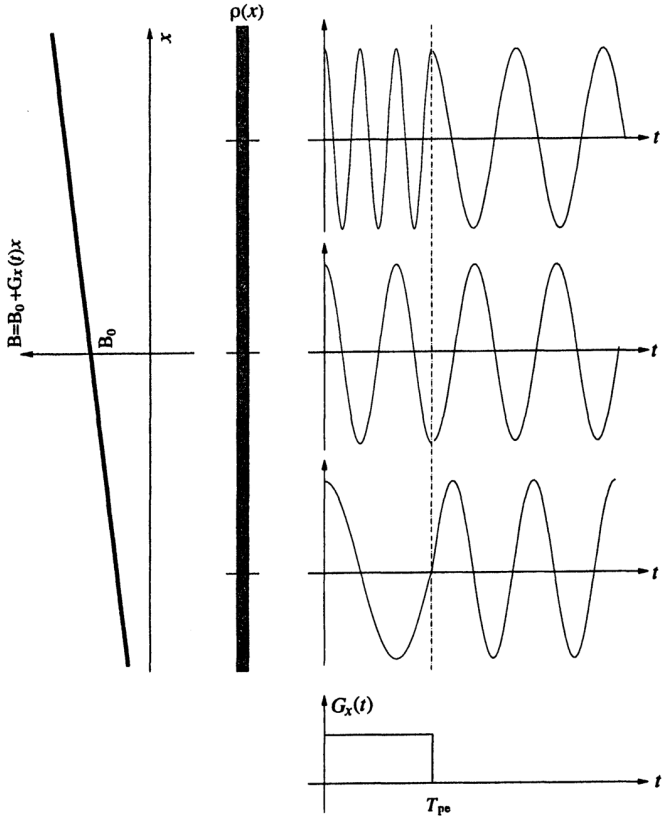


Figure 5.9 Phase-encoded signals from a one-dimensional object. Note that phase encoding is achieved by pre-frequency encoding the signals for a short period of time T_{pe}.

k-space

Signal from entire object with phase evolution due to local field deviation

$$S(t) \propto \int_{object} M_{xy}(\vec{r},0)e^{-i\Delta\omega(\vec{r})t}d\vec{r}$$

Local field in the presence of of a gradient (x-direction)

$$\Delta\omega(\vec{r})t = x\gamma\int_0^t G_x(\tau)d\tau$$

Insert and substitute

$$S(t) \propto \int_{object} M_{xy}(\vec{r},0)e^{-ix\gamma\int_0^t G_x(\tau)d\tau}d\vec{r}$$

$$2\pi k_x$$

k-space coordinate

$$2\pi k(t) = \gamma\int_0^t G(\tau)d\tau$$

k-space encoding and Fourier transform

$$S(t) \propto \int_{object} M_{xy}(\vec{r},0) e^{-ix\gamma \int_0^t G_x(\tau)d\tau} d\vec{r}$$

$$2\pi k_x$$

$$FT[g(x)] = G(k) = \int_{-\infty}^{+\infty} g(x) e^{-2\pi i k x}\, dx$$

Frequency encoding

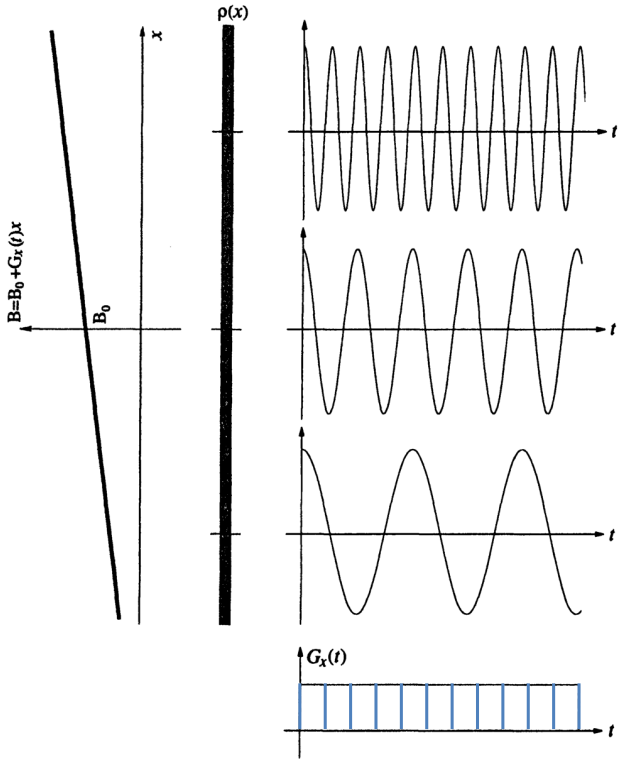


Figure 5.7 Localized signals from a hypothetical one-dimensional object in the presence of a frequency-encoding gradient.

Phase encoding

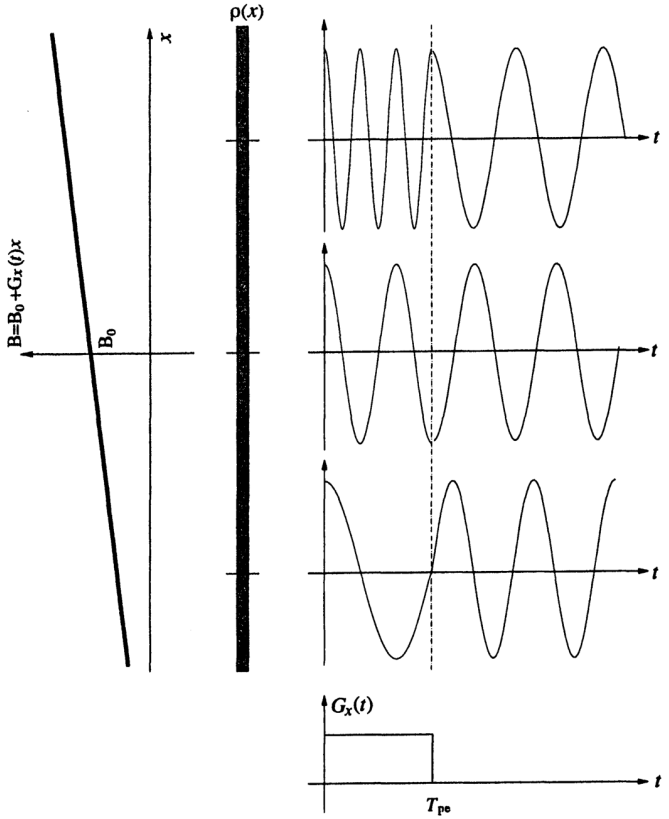
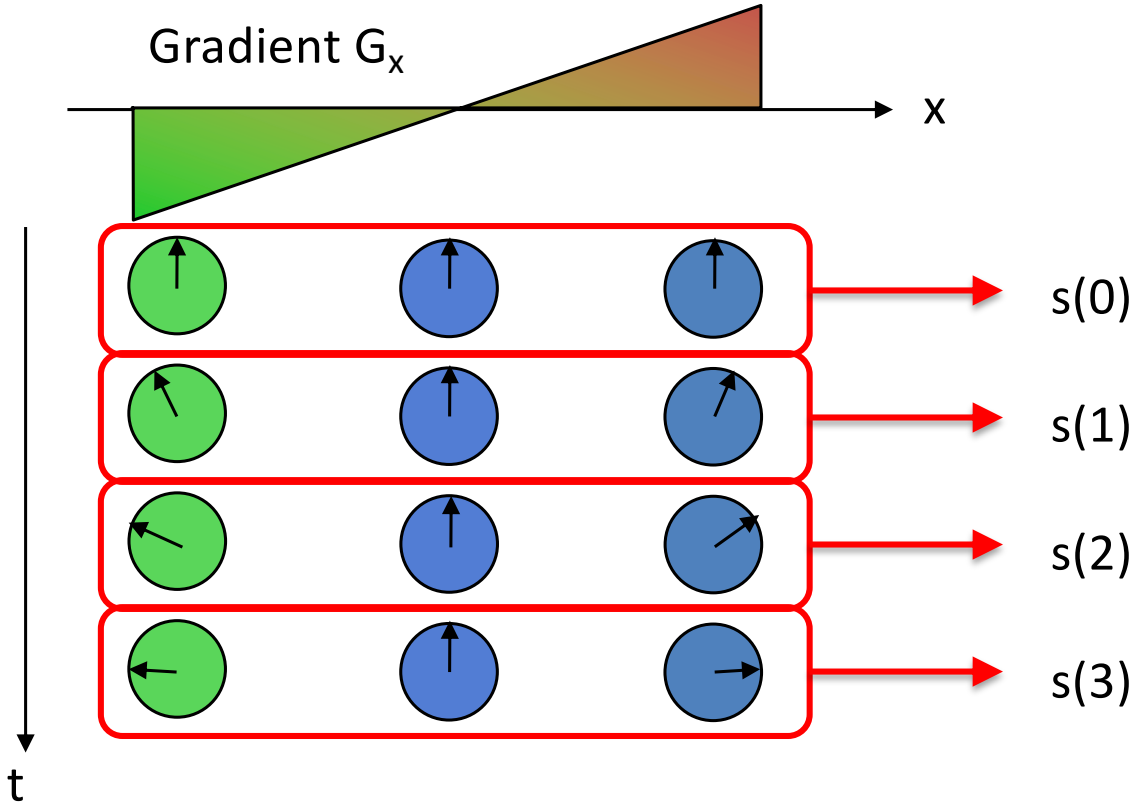
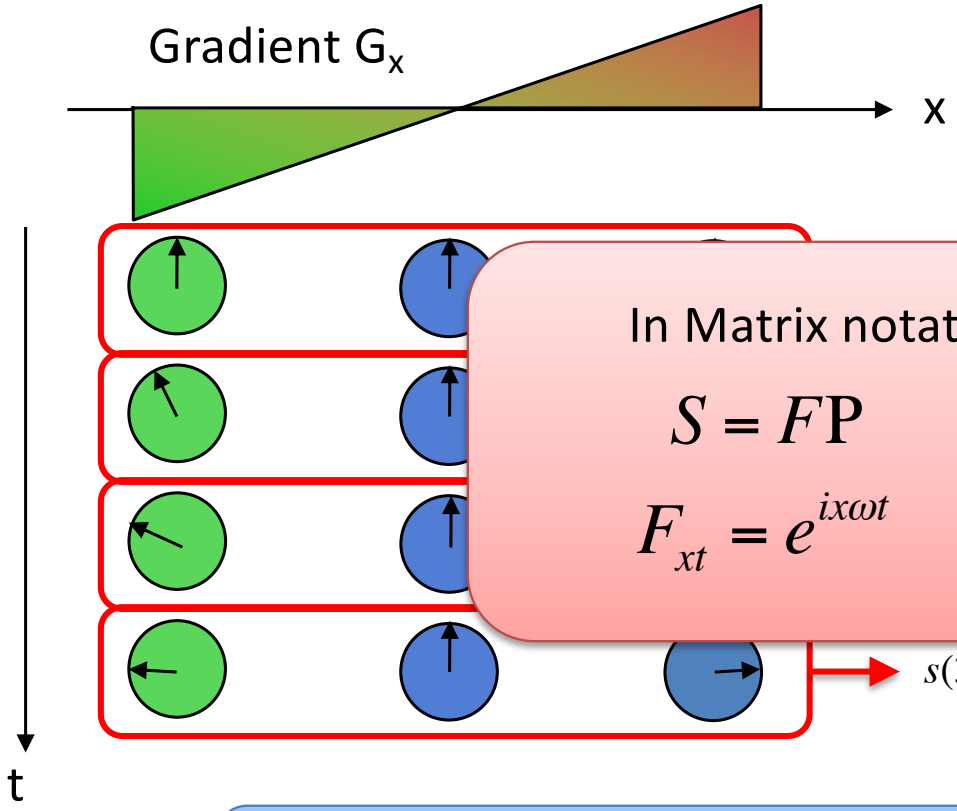


Figure 5.9 Phase-encoded signals from a one-dimensional object. Note that phase encoding is achieved by pre-frequency encoding the signals for a short period of time T_{pe}.

Signal in time domain



Signal in time domain



$$s(t) = \sum_x \rho(x) e^{i\psi(t,x)}$$

$$\psi(t,x) = x\omega t$$

$$s(0) = \rho(-1)e^{i\cdot -1\cdot\omega\cdot 0} + \rho(0)e^{i\cdot 0\cdot\omega\cdot 0} + \rho(1)e^{i\cdot 1\cdot\omega\cdot 0}$$

$$s(1) = \rho(-1)e^{i\cdot -1\cdot\omega\cdot 1} + \rho(0)e^{i\cdot 0\cdot\omega\cdot 1} + \rho(1)e^{i\cdot 1\cdot\omega\cdot 1}$$

$$s(2) = \rho(-1)e^{i\cdot -1\cdot\omega\cdot 2} + \rho(0)e^{i\cdot 0\cdot\omega\cdot 2} + \rho(1)e^{i\cdot 1\cdot\omega\cdot 2}$$

$$s(3) = \rho(-1)e^{i\cdot -1\cdot\omega\cdot 3} + \rho(0)e^{i\cdot 0\cdot\omega\cdot 3} + \rho(1)e^{i\cdot 1\cdot\omega\cdot 3}$$

In Matrix notation

$$S = F\mathrm{P}$$

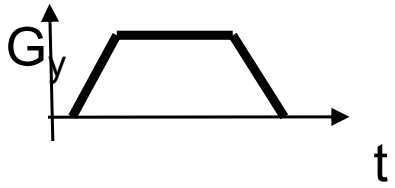
$$F_{xt} = e^{ix\omega t}$$

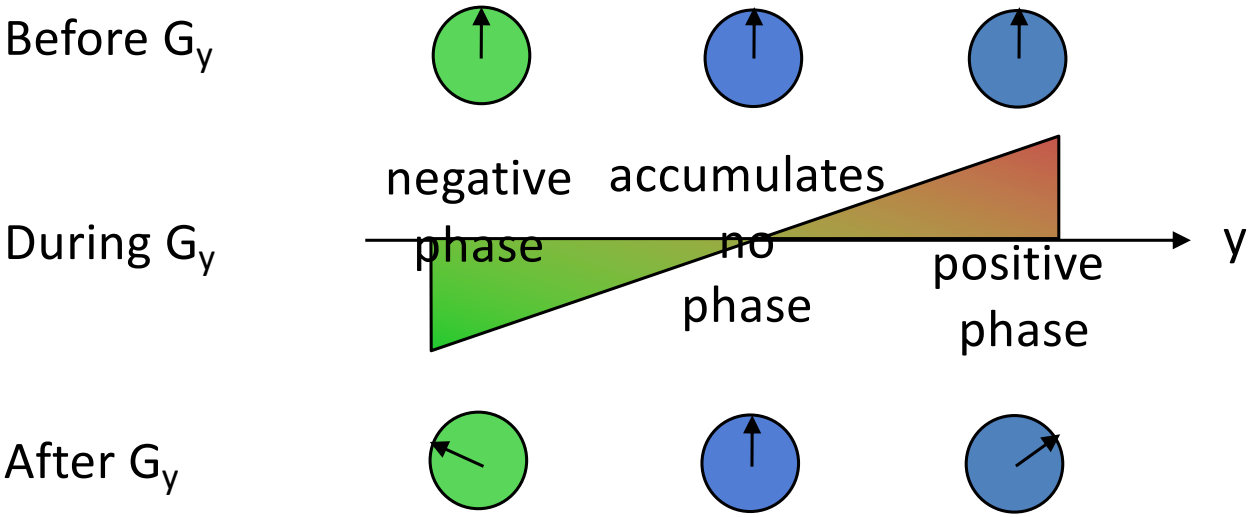
Reconstruction

$$\tilde{\mathrm{P}} = F^{-1}S$$

Phase encoding in the y direction

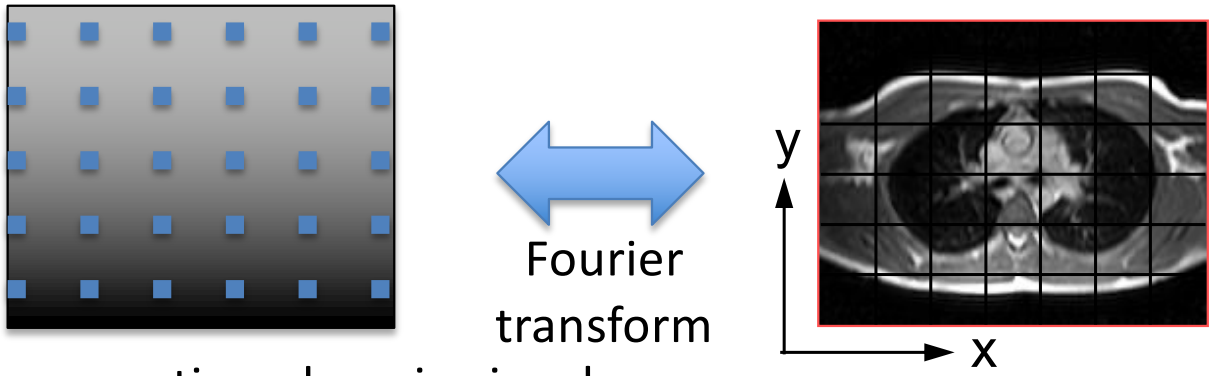
- Apply a short gradient lobe (on-off)





In-plane encoding in two directions

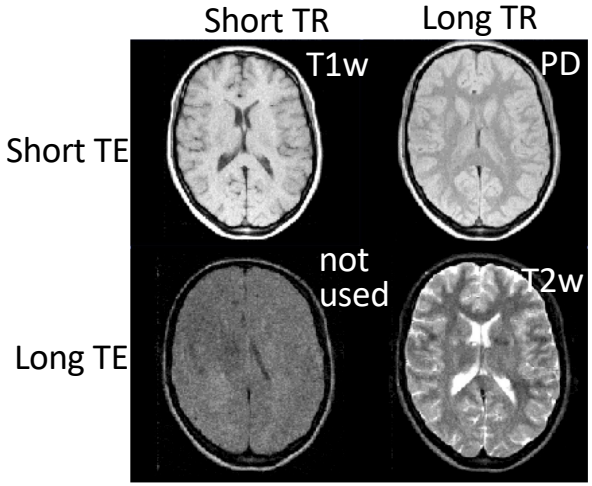
- Need all combinations to reconstruct an N x N pixel image
- x direction in one go (frequency encoding)
  - acquires N samples per readout
- y direction sequentially (phase encoding)
  - repeated N times, with varying strength of the G_y gradient pulse

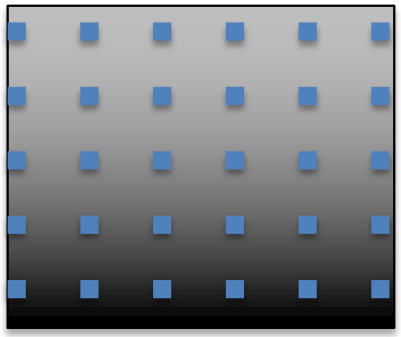


k-space – time domain signal

Scan time

- Need to collect N phase encodings
- Each TR, we can acquire N time domain samples (frequency encoding)
- For an image of N x N, the scan time becomes N x TR
- TR is chosen for the appropriate contrast
  - 256 x 256 with TR of 100 ms => 25.6 s scan time for one slice





k-space – time domain signal

Summary

- Slice direction is encoded during excitation
  - z gradient on -> linear frequency in z-direction
  - RF pulse with suitable bandwidth excites only spins within a band of Larmor frequencies
- In-plane encoding during signal reception
  - Frequency encoding (x) direction, gradient on during readout
    - Different positions have different frequency
  - Phase encoding (y) direction, gradient lobe of varying strength before each readout
    - Different positions have different phase
  - Reconstructed using Fourier transform
- MR measures the complex-valued Fourier transform of the image

1. Vad står förkortningen MR för?
☐ Magnetröntgen
☐ Mjukdelsröntgen
☐ Magnetresonans

2. Från vad kommer MR-signalen?
☐ Från vatten
☐ Från alla atomkärnor
☐ Från väteatomkärnan

3. Vilken av följande komponenter behövs för att ta bilder med en magnetkamera?
☐ En supraledande magnet
☐ Gradientspolar
☐ Ytspolar

4. Den kraftigaste kontrastmekanismen i MR bygger på skillnader i?
☐ T_1 och T_2
☐ Protondensitet
☐ Resonansfrekvens

5. Frekvensen på Larmor-precessionen beror på?
☐ Frekvensen på RF-sändaren
☐ T_1-relaxationen
☐ Magnetfältstyrkan och typ av atomkärna

6. Vilket påstående är sant?
☐ T_1-relaxationen är alltid snabbare än T_2-relaxationen
☐ T_2^* är alltid kortare än T_2
☐ Efter excitering avklingar signalen på grund av T_1-relaxationen

7. Gradienter används för?
☐ Frekvenskodning och faskodning
☐ Snittselektion

8. Vilket påstående om MR-signalen är sant?
☐ Bara magnetisering i samma riktning som huvudmagnetfältet kan detekteras
☐ MR-signalen är proportionell mot nettomagnetiseringen (M_0)
☐ Bara magnetisering i transversalplanet (xy-planet) kan detekteras

9. Vilket påstående är sant?
☐ Signalen innehåller bidrag från hela den valda skivan
☐ Intensiteten i en bildpixel är densamma som motsvarande pixel i *k*-space
☐ I varje faskodningssteg kodas en rad pixlar av MR-bilden

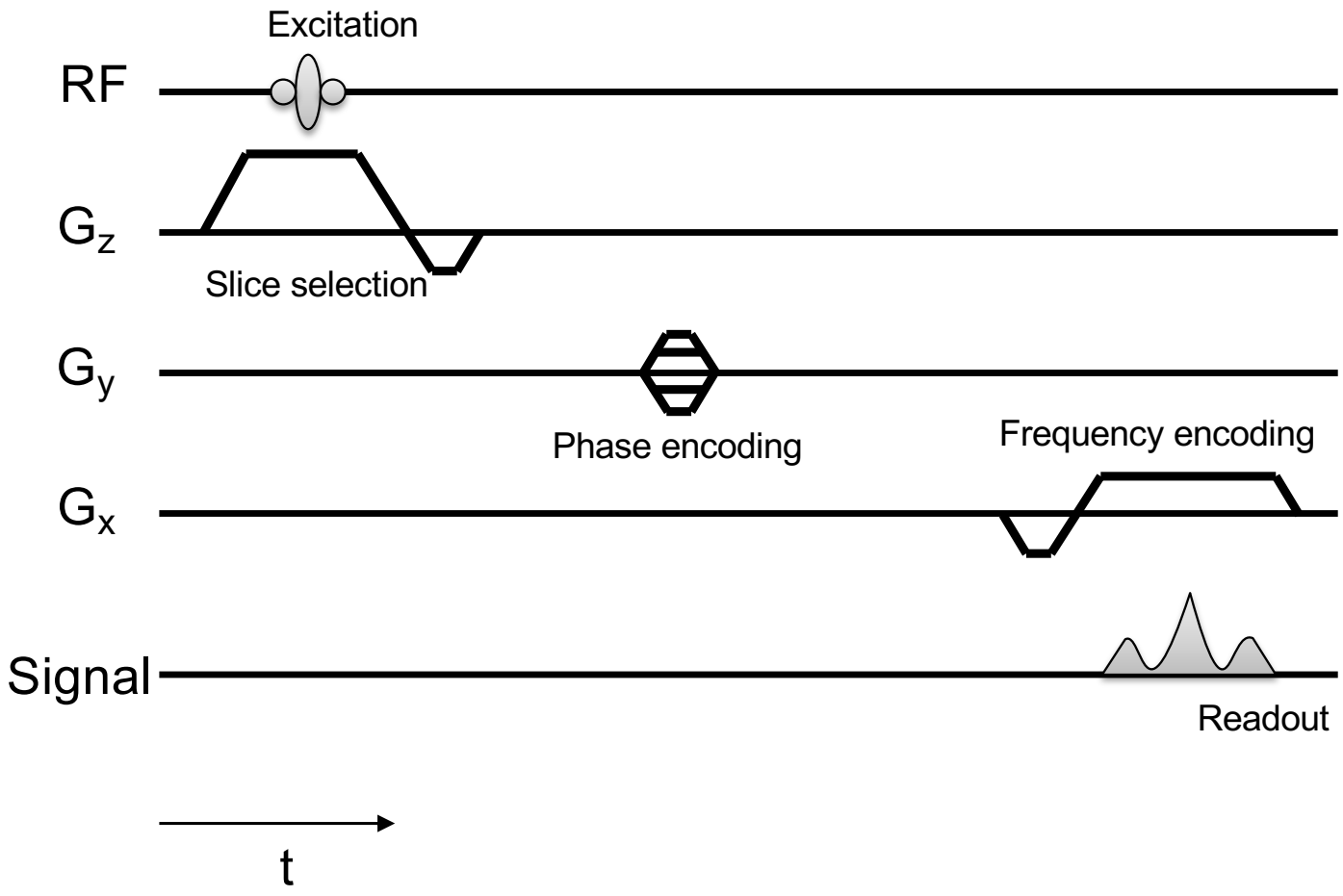
10. En vävnad med lång T_1 och lång T_2 är ljus
☐ På en T_1-viktad bild
☐ På en T_2-viktad bild

Part III

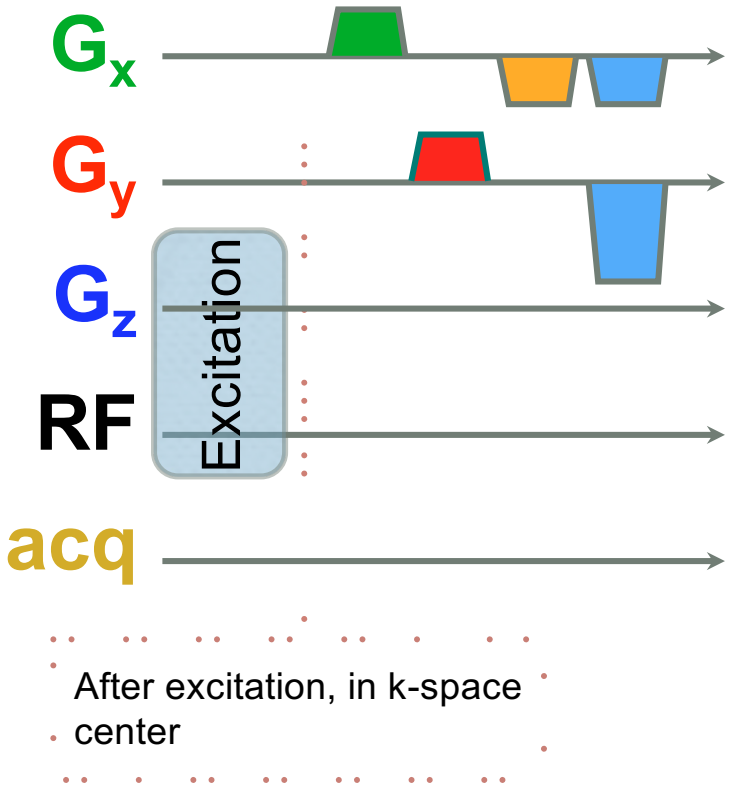
MRI is versatile

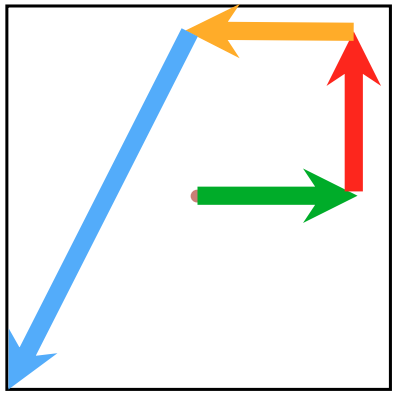
- Means of manipulation
  - RF pulses
    - Only on-resonance spins get excited
    - Arbitrary flip angle
    - Prepulses
  - Gradients
    - Slice selection: change resonance frequency spatially
    - Spatial encoding: position gets encoded in phase
  - Time
    - Relaxation, transverse (T_2) and longitudinal (T_1)
  - Motion
    - Phase shifts and dephasing during gradients

Pulse sequences



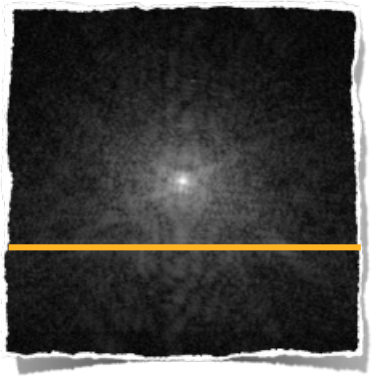
Gradient area & k-space

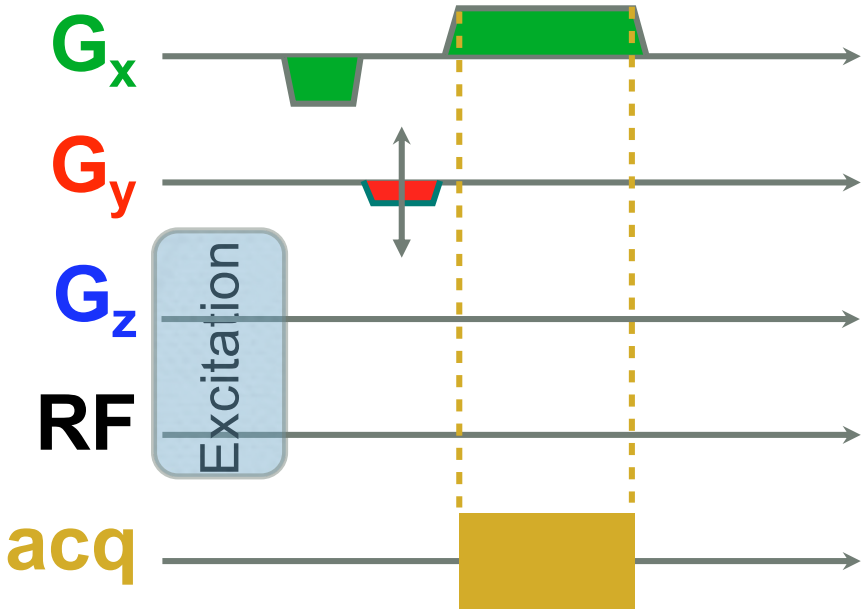


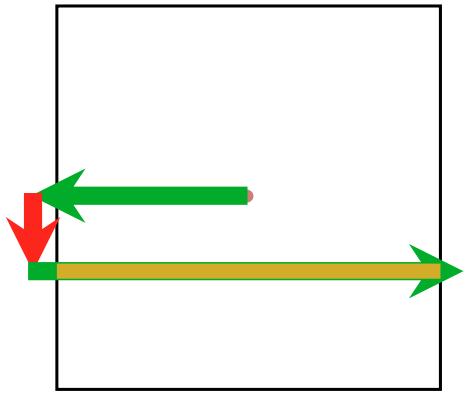


k-space [1/m]

Gradient echo







k-space [1/m]

Linear left-right k-space filling



Linear top-down k-space filling



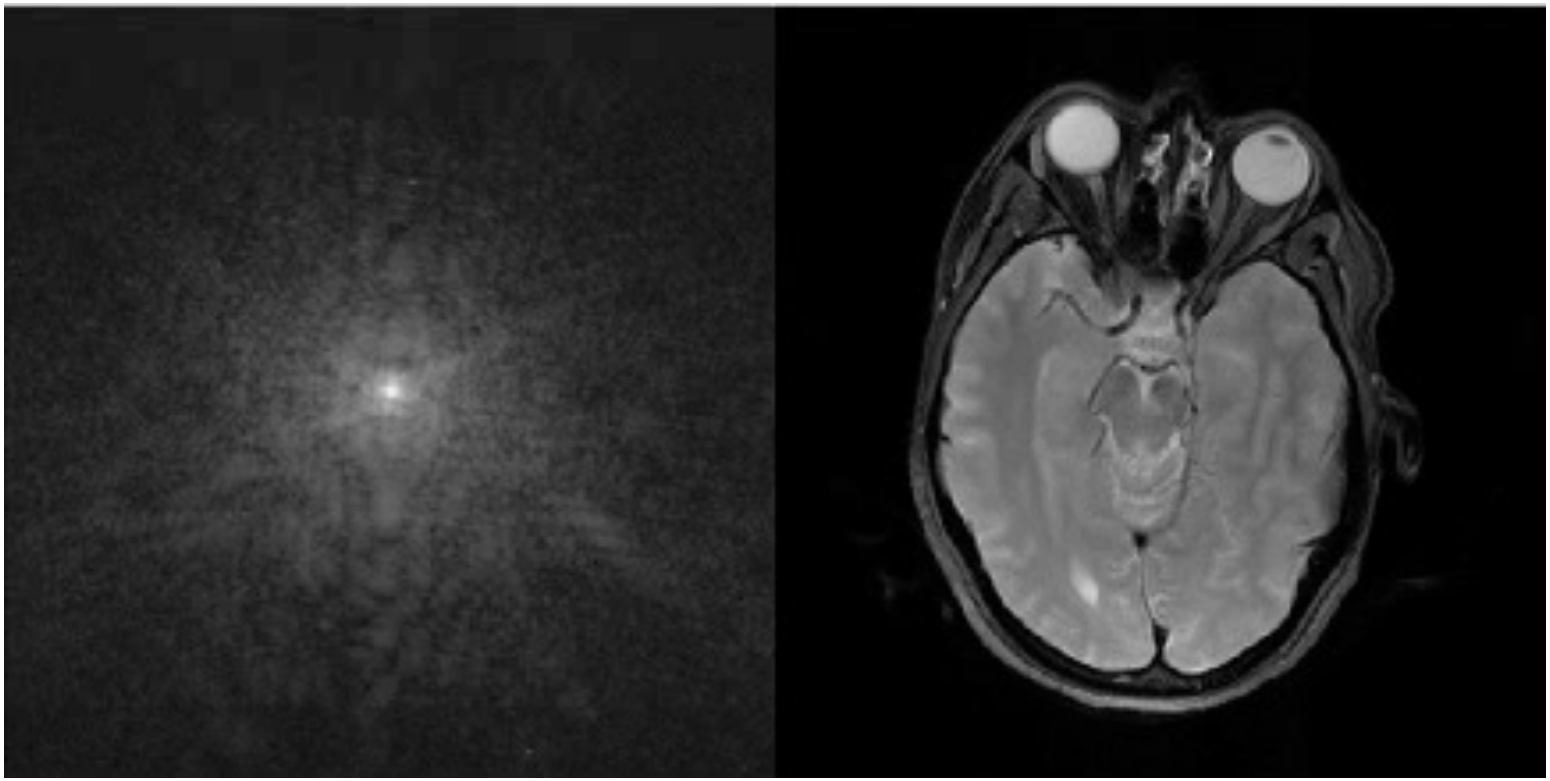
Center-out vertical k-space filling



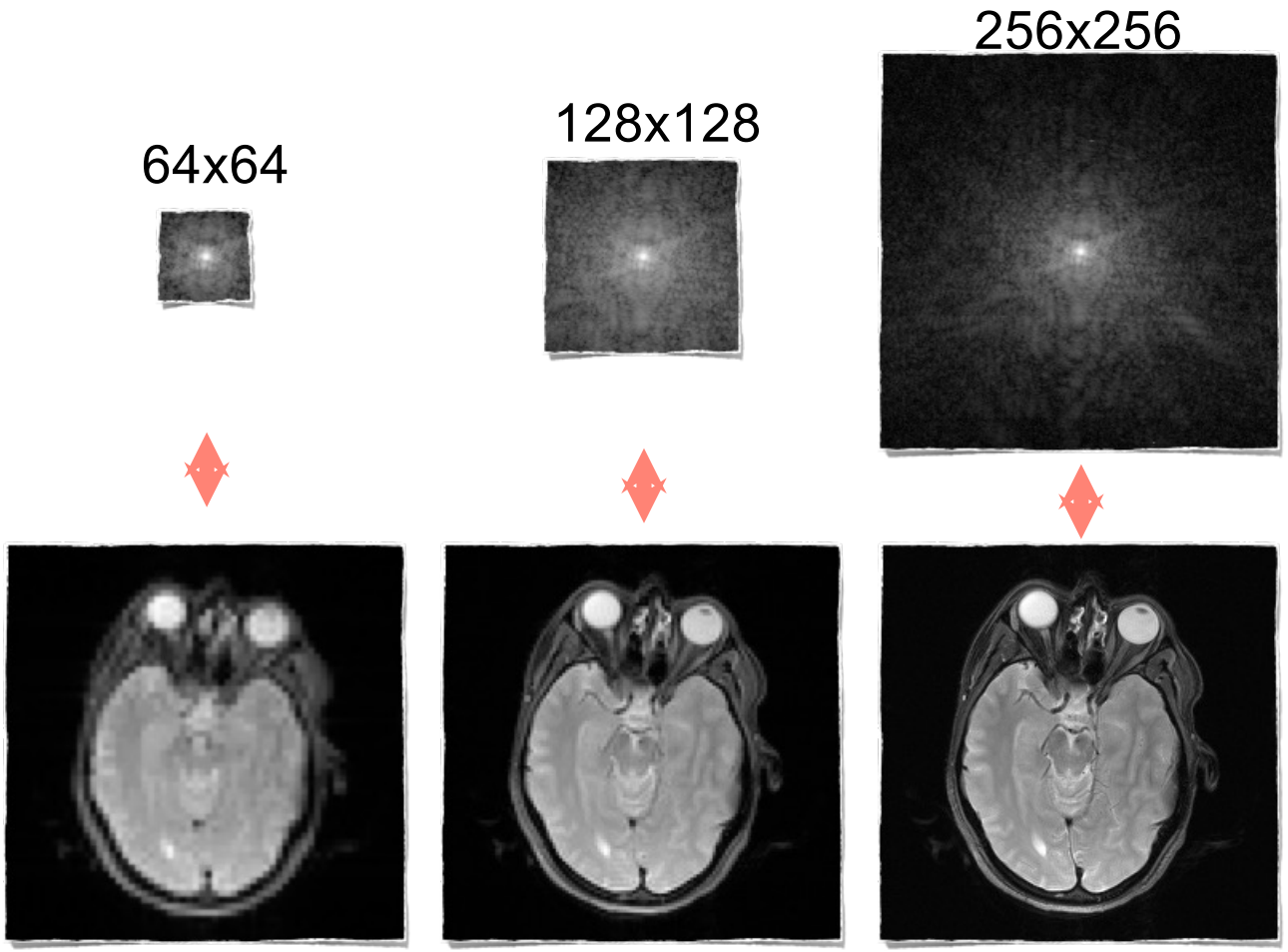
From 2x2 to 256x256



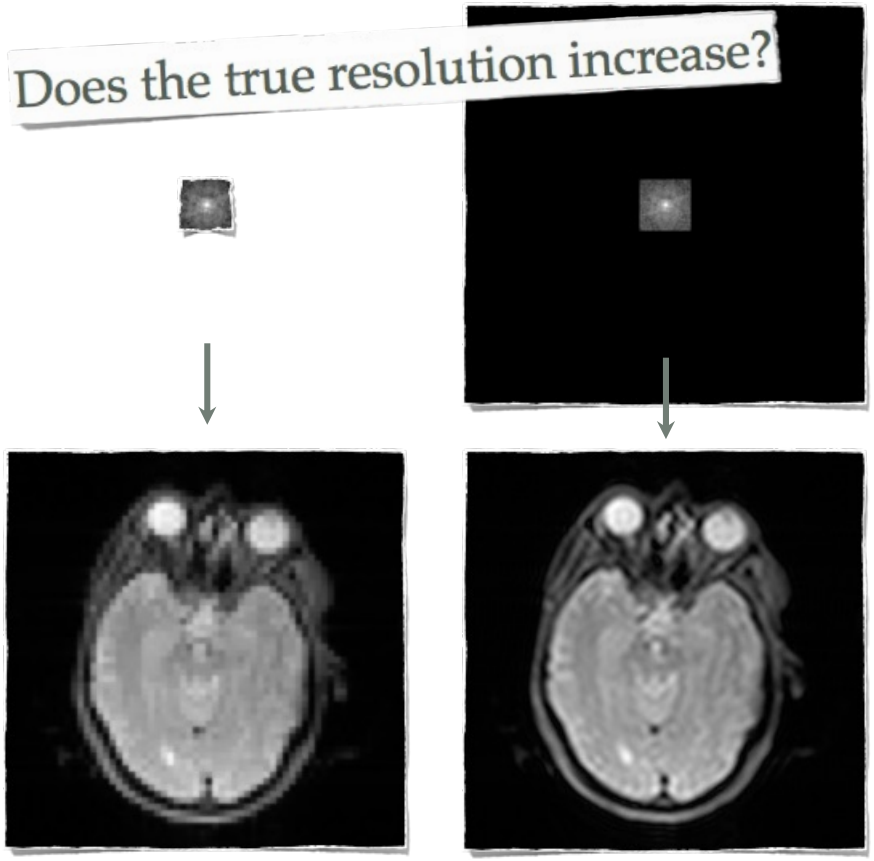
Removal: center-out



k-space size



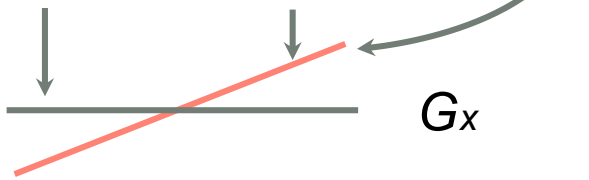
Zero-filling 64x64 -> 512x512



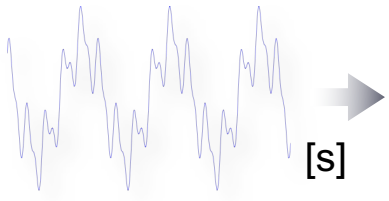
Signal, k-space, imag

With a gradient G_x
we map frequencies [1/s] to distances [m]

Low freq. [1/s] High freq. [1/s]



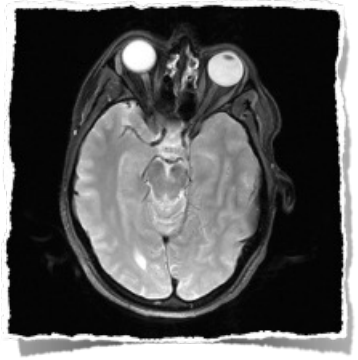
G_x



[s]



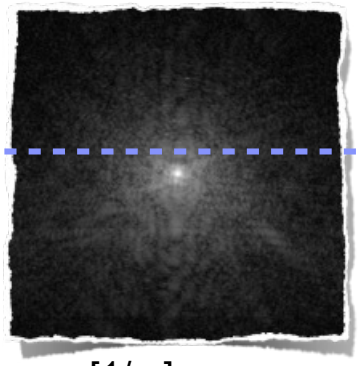
[m]



[m]

Fourier transform
(units are inverted!)



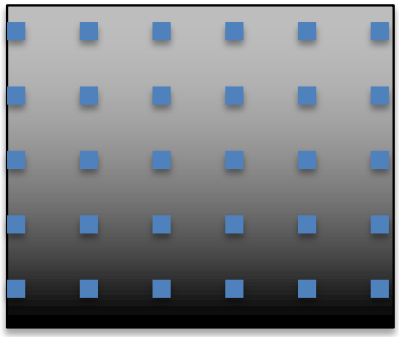


[1/m]

[1/m]
(and "[s]")

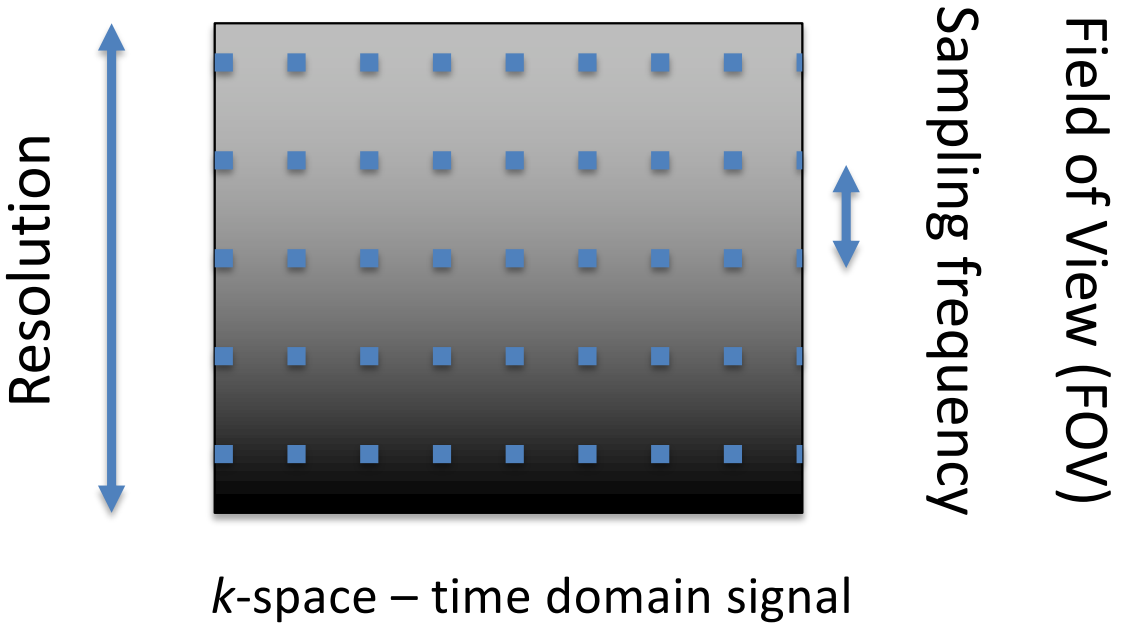
Scan time

- 256 x 256 with TR of 100 ms => 25.6 s scan time for one slice
- Shorten scan time
  - Reduce TR
    - Changes contrast
  - Reduce number of lines in k-space
    - Reduces resolution
    - or reduces Field Of View (FOV)

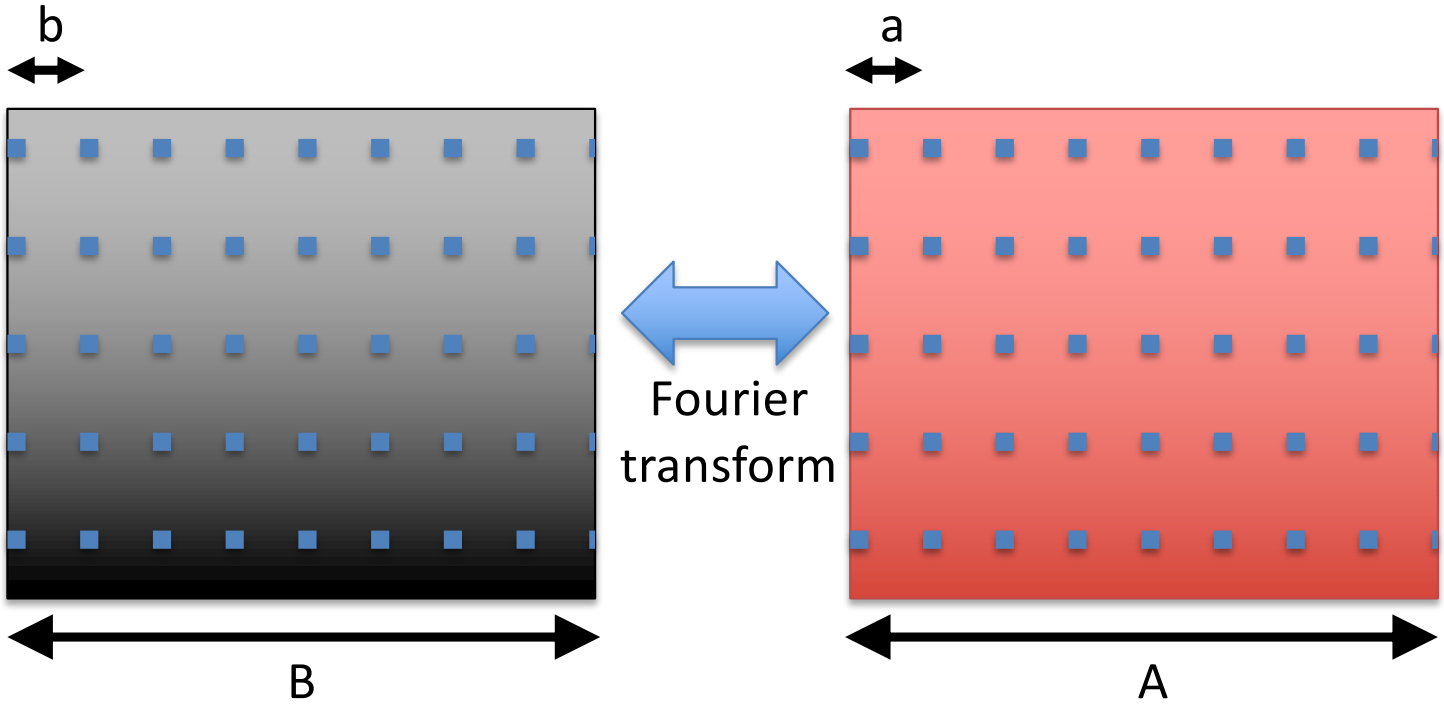


k-space – time domain signal

k-space matrix properties



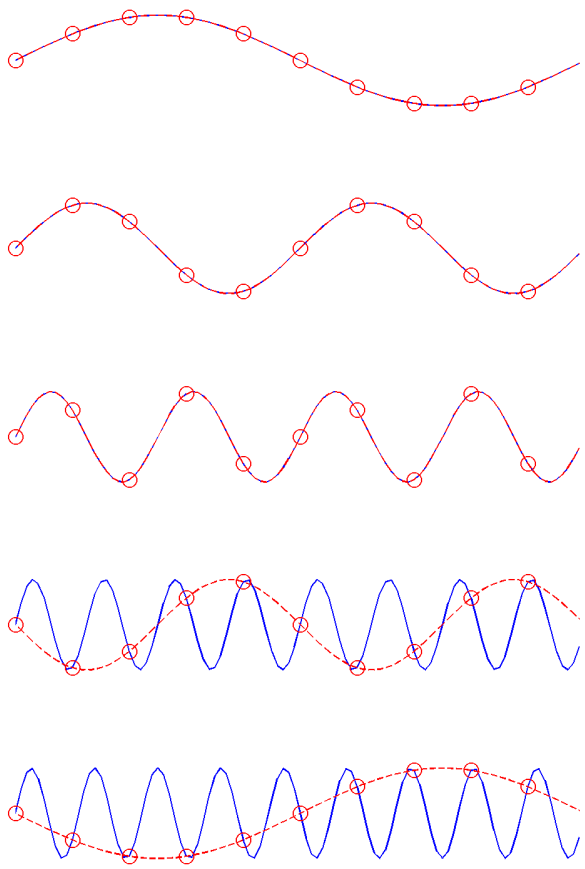
k-space relationships



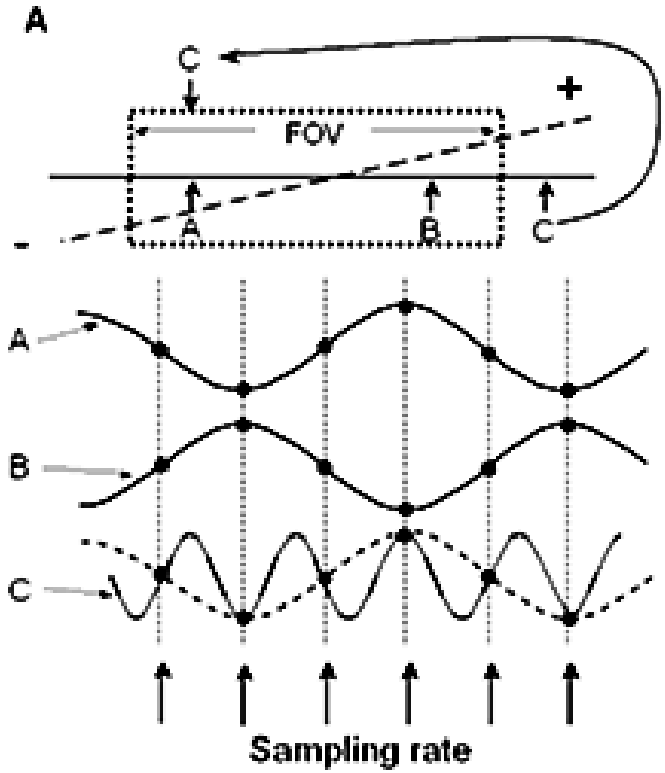
a ~ 1 / B
b ~ 1 / A

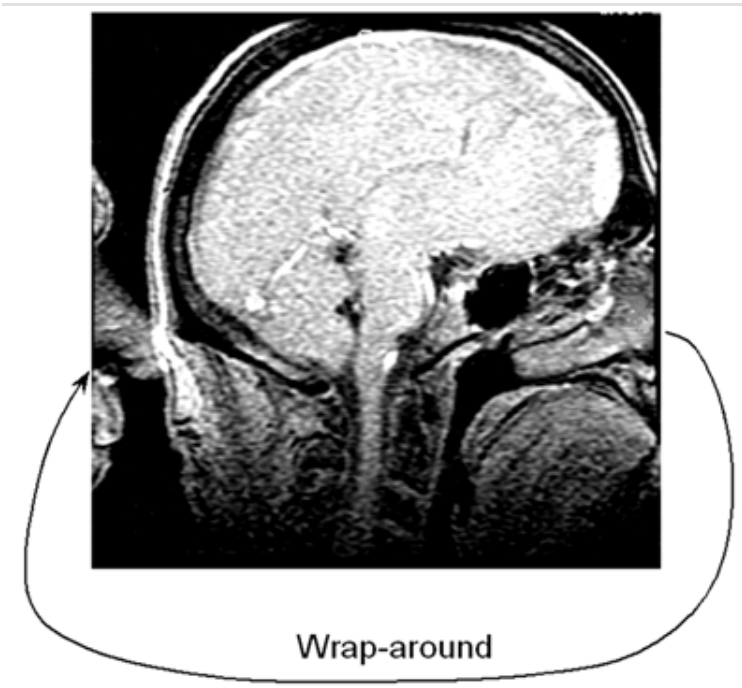
Fold-over artifacts

- Frequency determination relies on sufficient sampling frequency
- Can only resolve up to Nyquist frequency
- Nyquist frequency is half of sampling frequency
  - Corresponds to two samples per period
- If Nyquist sampling criterion is not fulfilled
  “aliasing” occurs



Fold-over artifacts

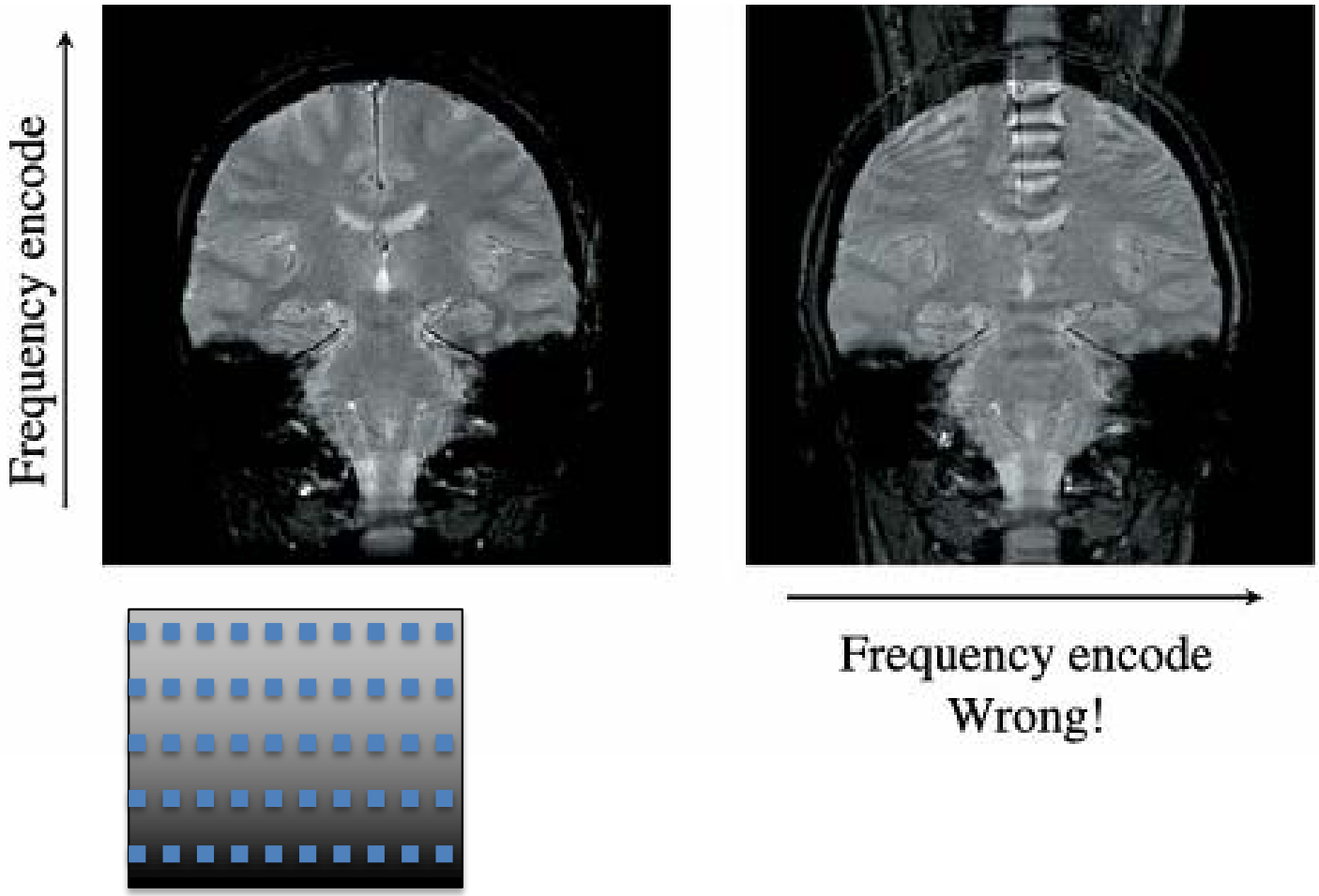




Fold-over artifacts

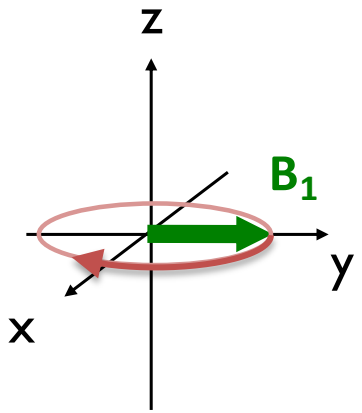


Importance of readout direction

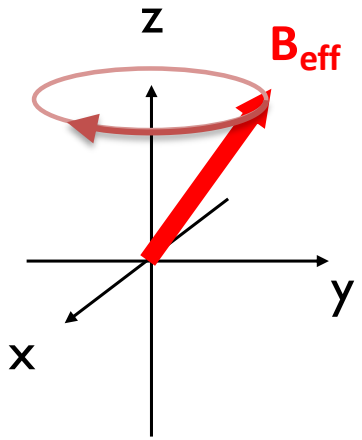


RF pulses

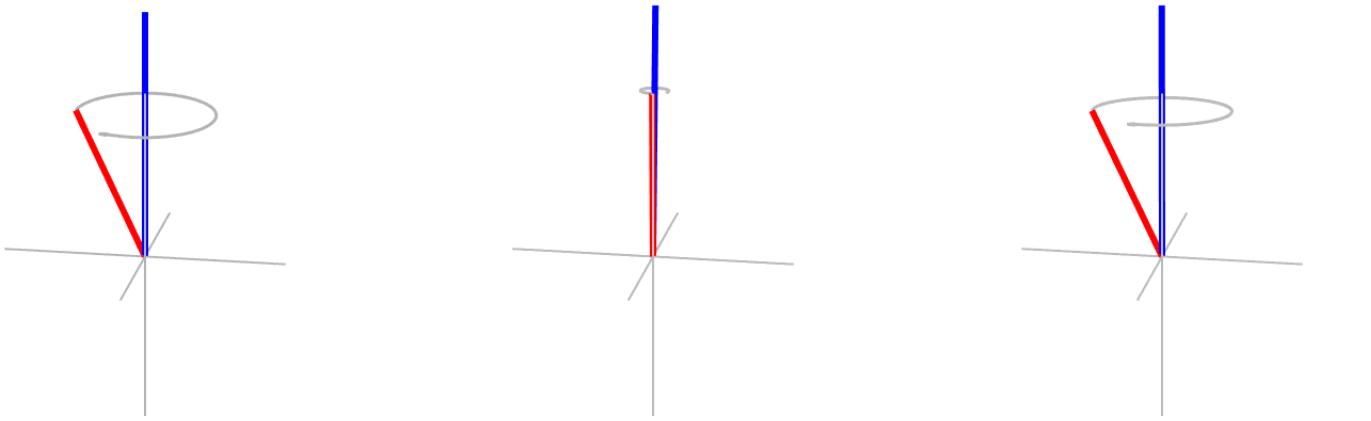
- Rotating magnetic field
  - Not radio waves
  - H field, ideally no E field



- $B_{eff} = B_0 + B_1$
  - B_0 ~ 1.5-3 T, 0 Hz
  - B_1 ~ 30 µT, 64-128 MHz



RF pulses - forced precession



No RF pulse

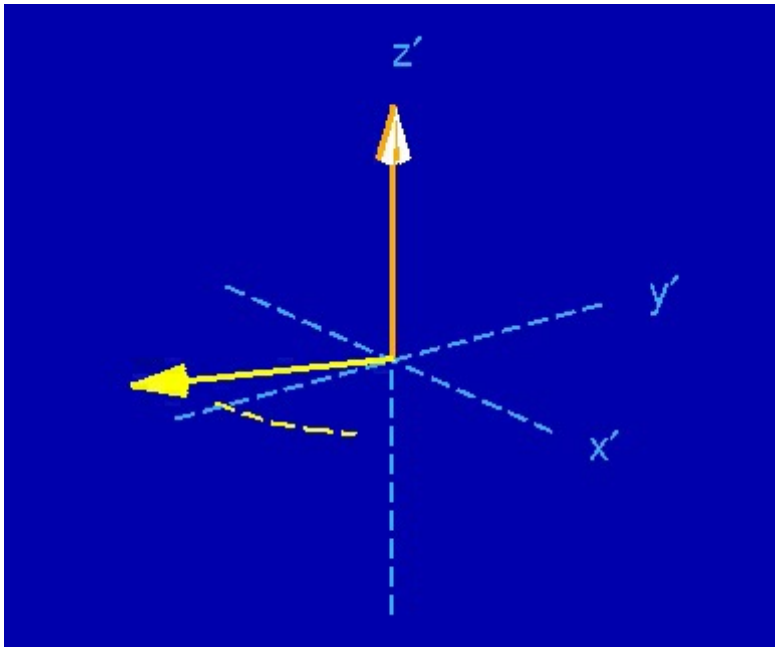
RF on resonance

RF off resonance

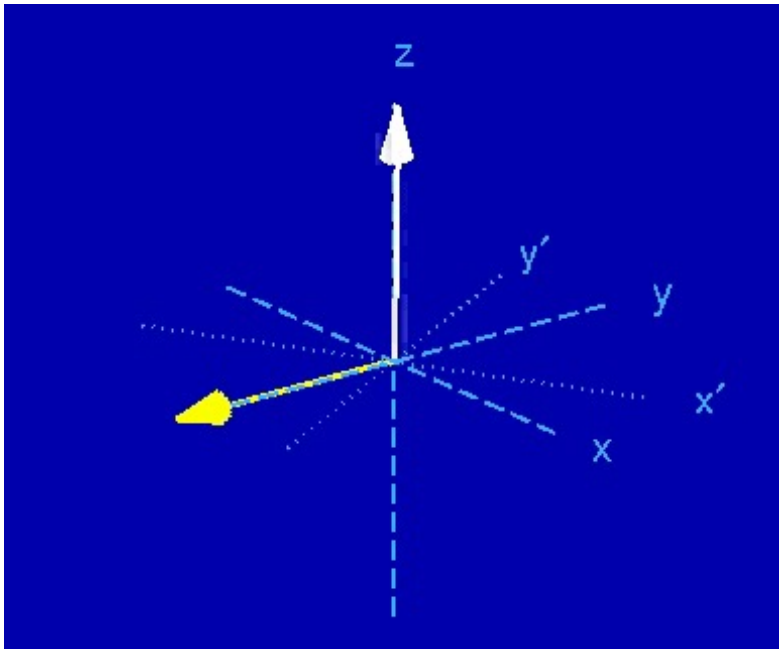
Magnetization

Effective B1

Rotating frame of reference



Static frame of reference



Rotating frame of reference

Movies from http://mrsrl.stanford.edu/~brian/intromr/ (Brian Hargreaves)

Simulation

- http://www.drcmr.dk/bloch
- Bloch equations (after Felix Bloch); differential equations in matrix form that describe precession, relaxation and other manipulations

$$\frac{dM_x(t)}{dt} = \gamma(\mathbf{M}(t) \times \mathbf{B}(t))_x - \frac{M_x(t)}{T_2}$$

$$\frac{dM_y(t)}{dt} = \gamma(\mathbf{M}(t) \times \mathbf{B}(t))_y - \frac{M_y(t)}{T_2}$$

$$\frac{dM_z(t)}{dt} = \gamma(\mathbf{M}(t) \times \mathbf{B}(t))_z - \frac{M_z(t) - M_0}{T_1}$$

Chemical shift

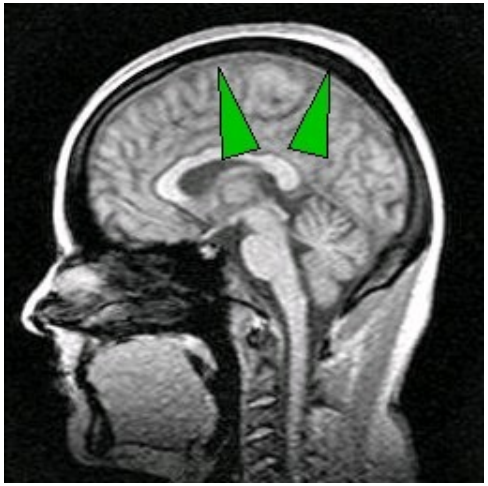
- Electrons shield the magnetic field
- Electron distribution depends on molecular structure
- Local magnetic field will be different for different molecules

- Chemical shift
- Can be used to discriminate different molecules
  - Despite that the signal is from the same nuclei
- Most apparent with water and fat
  - ~150 Hz/T difference

Water-fat shift

- Water-based tissues and fat-based tissues have different chemical shift
- The position encoding relies on frequency differences caused by the gradient
- Additional frequency differences due to chemical shift result in position decoding errors -> fat signal is shifted in the image



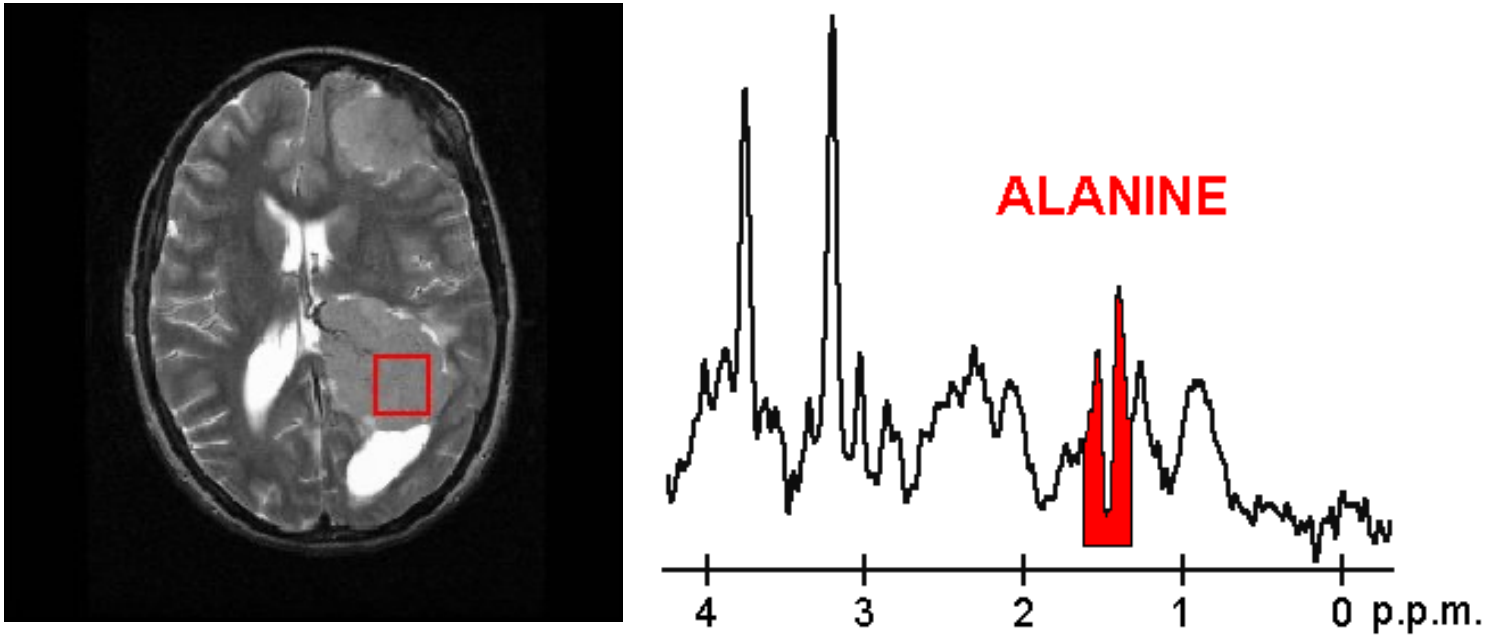


mr-tip.com

MR Spectroscopy

- In spatial encoding, a gradient was applied during readout
  - Reconstruct different positions depending on frequency
- MR spectroscopy
  - Readout without applied gradient
  - Different molecules give signal with different Larmor frequency
  - Similar reconstruction, but with a frequency dimension instead of a spatial dimension
- Spatial localization of MR spectroscopy requires extra effort
  - Single-voxel reasonably simple
  - MR spectroscopic imaging, requires more acquisition time

MR Spectroscopy

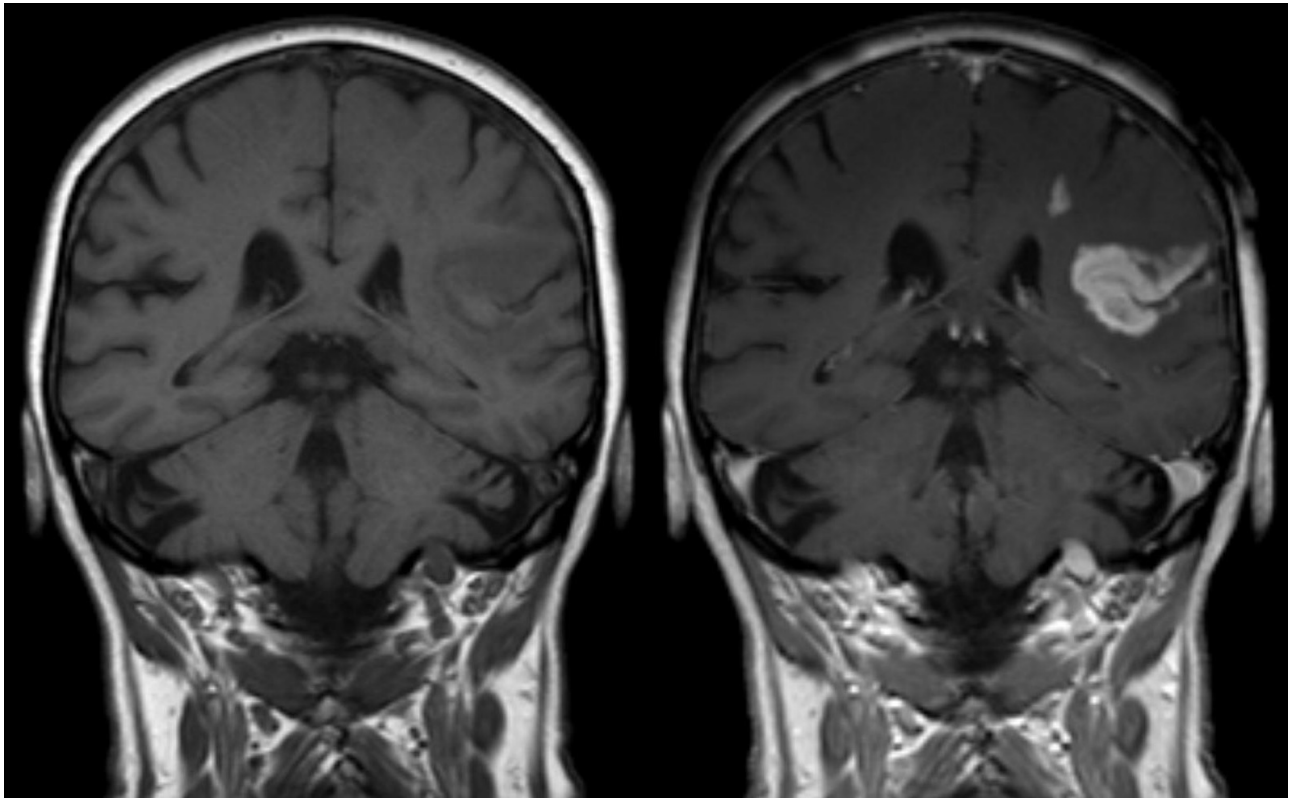


The University of Hull Centre for Magnetic Resonance Investigations (http://www.hull.ac.uk/mri).

Contrast agents

- Most MRI contrast agents don’t give signal themselves
- Affects T_1 and T_2
- Lower dose than X-ray / CT contrast agents
- Gadolinium-based
  - Paramagnetic
  - Rare-earth metal
  - Toxic, but chelated in practice
  - Reduces T_1
- Iron oxide
  - Reduces T_2

Contrast agents



Defect of the blood-brain barrier after stroke shown in MRI. T1-weighted images, left image without right image with contrast medium administration.

The insides of an MRI system

Superconducting magnet,
cooling system (liquid N, He)

Gradient coils
sealed in epoxy

Integrated
RF coil

Electronics, cooling systems,
and more



Summary

- Some nuclei have spin

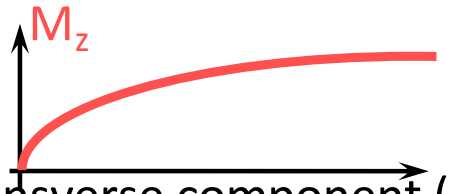


  - Split energy level in presence of magnetic field
  - Energy difference results in a Larmor frequency precession
  - Larmor frequency *proportional to magnetic field* (^{1}H, 1.5 T => 64 MHz)
  - Population difference very small (10^{-5})

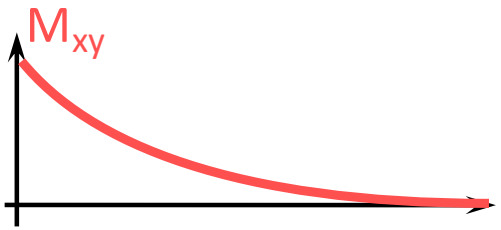
- Net magnetization vector M
  - A group of spins have a net magnetization vector
  - Can be measured when flipped to the transversal plane (M_{xy})
    - Complex signal with magnitude and phase
  - Relaxes back to thermal equilibrium (M_0 along B_0 direction)

Summary, relaxation

- T_1: Longitudinal relaxation, the net magnetization vector returns to thermal equilibrium along B_0

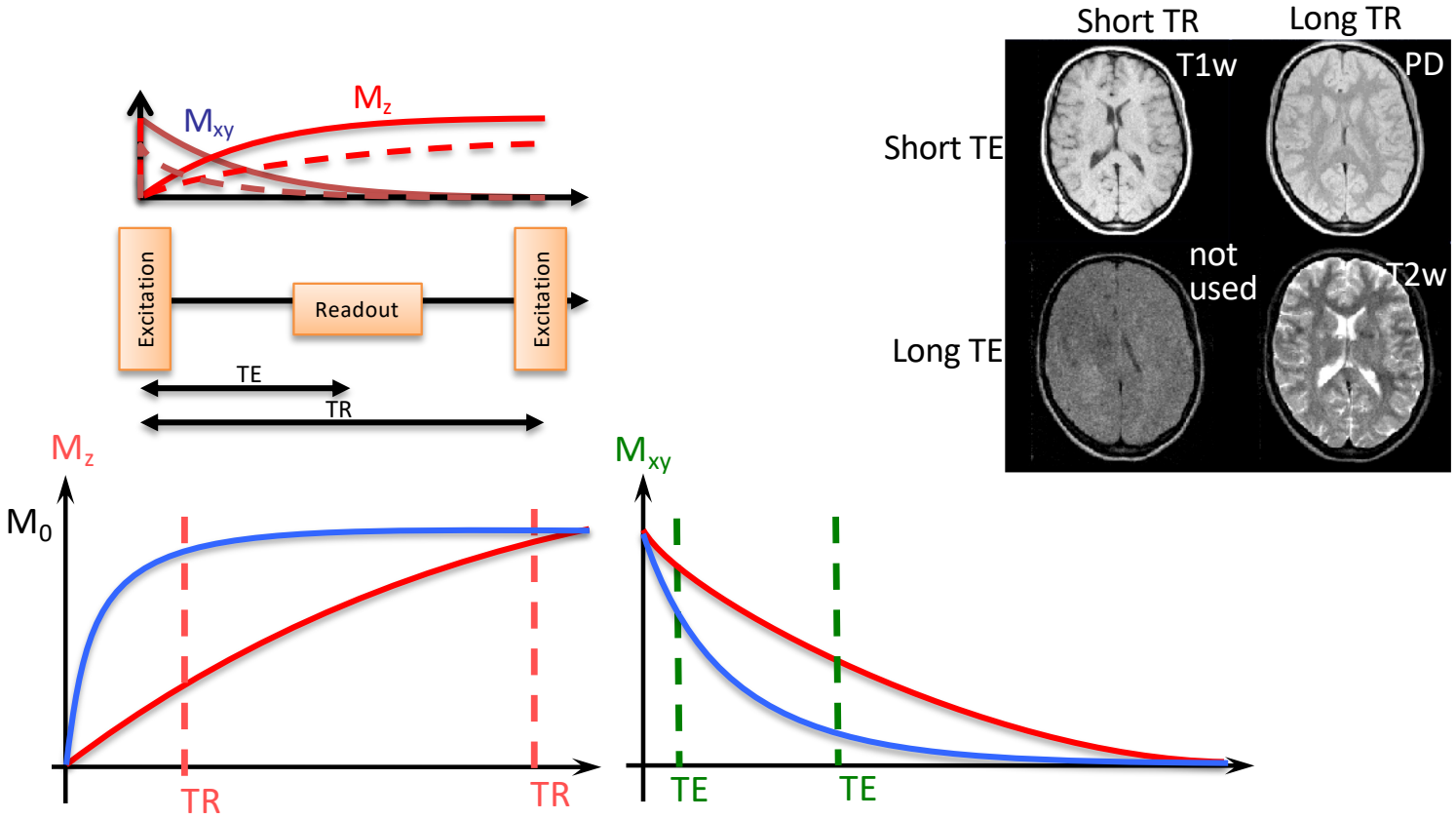


- T_2 and T_2^*: due to varying Larmor frequency, the transverse component (Mxy) of the net magnetization dephases and signal vanishes
  - T_2 is irreversible because they are random in time
  - T_2^* is the apparent T_2 when not using 180 degree refocusing pulses
  - The temporally static variations are refocused using a 180 degree pulse
- $T_2^* < T_2 < T_1$

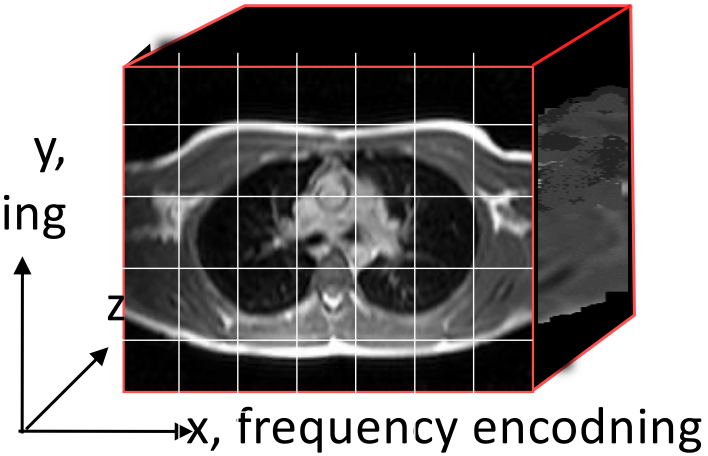


Summary, image contrast

- By choosing TE and TR appropriately, a desired image contrast can be obtained



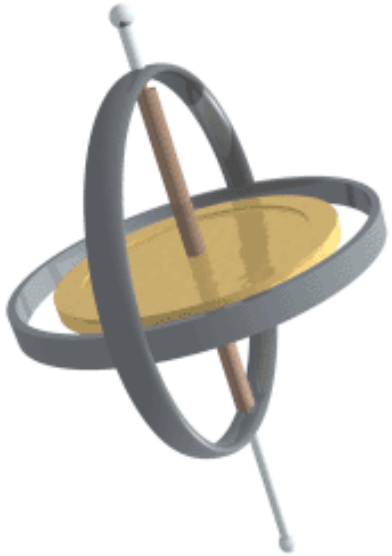
Summary, image localization



- Slice is selected during excitation
  - Gradient during RF pulse
  - Only spins within the bandwidth are excited
- Frequency encoding (x)
  - Gradient during signal reception (one TR)
  - Signal with frequency depending on position is emitted
  - Fourier transform (“spectrum analyzer”)
- Phase encoding (y)
  - Gradient lobe with varying strength before readout
  - Needs to be repeated for all combinations (N x TR)
  - Fourier transform for reconstruction

Acknowledgements

- Karin Markenroth Bloch, Lund (Slides)
- Andreas Sigfridsson, KI (Slides)



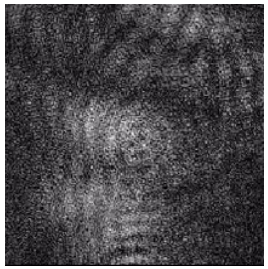
The next step

- Course in Magnetic Resonance Imaging
  - HL2011, autumn, 6 HP
- Master projects
  - MR Fingerprinting, neural network reconstructions alexander.fyrdahl@ki.se

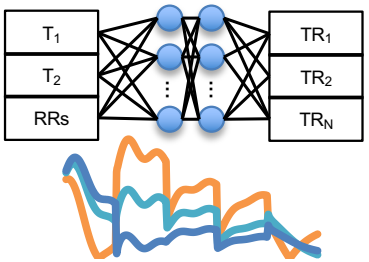
Spiral readout



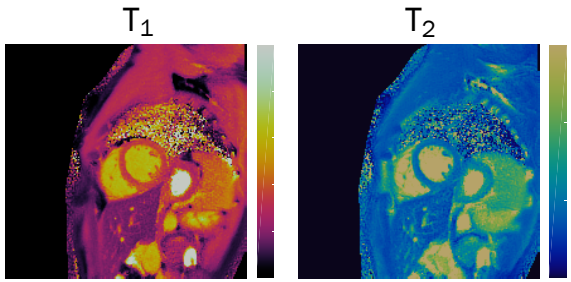
k-Space



MRF Dictionary



Parametric maps



Book chapter:

Fyrdahl A et. al. Magnetic Resonance Fingerprinting: The Role of Artificial Intelliger